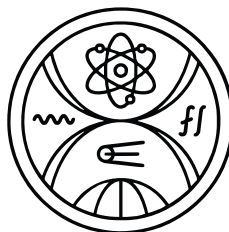
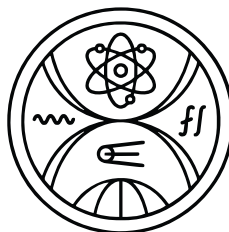


COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS



NEW APPROACHES TO NOWHERE-ZERO FLOW PROBLEMS
DIPLOMA THESIS

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FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS



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DIPLOMA THESIS

Study Programme: Computer Science
Field of Study: Computer Science
Department: Department of Computer Science
Supervisor: Mgr. Jozef Rajník, PhD.

Bratislava, 2026
Bc. Lukáš Gáborik



Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

ZADANIE ZÁVEREČNEJ PRÁCE

Meno a priezvisko študenta: Bc. Lukáš Gáborik
Študijný program: informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor: informatika
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Názov: New approaches to nowhere-zero flow problems
Nové prístupy k problémom o nikde-nulových tokoch

Anotácia: Táto práca nadväzuje na výsledky bakalárskej práce rovnakého autora. Novo zavedený pojem viacrozmerých Manhattanských a Čebyševovských tokov ponecháva priestor pre ďalší výskum, ako napríklad hľadanie dolných odhadov pre tokové čísla. Jedným z hlavných výsledkov spomínanej bakalárskej práce je predstavenie hypotézy tvrdiacej, že každý bezmostový graf pripúšťa (1,2)-cirkulačnú dekompozíciu, t. j. 2-cirkuláciu a 4-cirkuláciu takú, že zakaždým keď je 2-cirkulácia nulová na nejakej hrane, tak 4-cirkulácia nemôže nadobúdať 0, +1 alebo -1. To ponúka bohatý priestor na výskum vrátane rôznych zovšeobecnení, v ktorých kladieme ďalšie požiadavky na tokové hodnoty.

Cieľ:

1. Dokázať netriviálne dolné odhady pre dvojrozmerné Čebyševovské tokové číslo grafu.
2. Preskúmať možné spôsoby, ako dokázať hypotézu, že každý bezmostový graf pripúšťa (1, 2)-cirkulačnú dekompozíciu, a iné súvisiace hypotézy. Dokázať túto hypotézu pre niektoré nekonečné triedy snarkov, prípadne pre niektoré snarky, ktoré majú ďaleko od toho, aby boli zafarbitelné (napr. s nepárnosťou 2, indexom perfektného párenia 4, ...).
3. Preskúmať ďalšie zovšeobecnenia tokov potenciálne užitočné v kontexte vyššie uvedenej hypotézy.

Vedúci: Mgr. Jozef Rajník, PhD.
Katedra: FMFI.KI - Katedra informatiky
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Dátum zadania: 21.11.2024

Dátum schválenia: 05.12.2024

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.....
študent

.....
vedúci práce



THESIS ASSIGNMENT

Name and Surname: Bc. Lukáš Gáborik
Study programme: Computer Science (Single degree study, master II. deg., full time form)
Field of Study: Computer Science
Type of Thesis: Diploma Thesis
Language of Thesis: English
Secondary language: Slovak

Title: New approaches to nowhere-zero flow problems

Annotation: This work builds on the results of the bachelor thesis of the same author. The newly introduced notion of multidimensional Manhattan and Chebyshev flows still leaves some possibilities for further research like finding lower bounds on flow numbers. One of the main results of the mentioned bachelor thesis is the introduction of the conjecture asserting that each bridgeless graph admits a $(1,2)$ -circulation decomposition, that is a 2-circulation and a 4-circulation such that whenever the 2-circulation is zero on any edge, the 4-circulations can not attain 0, +1 or -1. This offers wide possibilities of exploration including various generalisations where further requirements are posed on the flow values.

Aim:

1. Prove nontrivial lower bounds on 2-dimensional Chebyshev flow number of a graph.
2. Explore possible ways of proving the conjecture that each bridgeless graph admits a $(1, 2)$ -circulation decomposition, and other related conjectures. Prove this conjecture for some infinite families of snarks, eventually for some snarks that are far from being colourable (e.g. with oddness 2, perfect matching index 4, ...)
3. Research other generalisation of flows potentially useful in the context of the abovementioned conjecture.

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Abstrakt

TODO

Klíčové slová: TODO

Abstract

A nowhere-zero k -flow is a mapping of values $\pm 1, \pm 2, \dots, \pm(k-1)$ to darts of a graph satisfying Kirchhoff's law – the inflow is the same as the outflow – for every vertex of the graph. It is conjectured [34] that each bridgeless graph has a nowhere-zero 5-flow. Almost three decades later, Seymour [29] made a general construction of nowhere-zero 6-flow from a 2-flow and a 3-flow. Although these two flows permit zeroes, no edge can have both of them equal to zero simultaneously. In a recent preprint [10], along with the introduction of two-dimensional Chebyshev nowhere-zero flows, the authors showed a similar construction of the nowhere-zero 5-flow from a 2-flow and a 4-flow. In addition to forbidding simultaneous zero values, the flow values $(0, 1)$ and $(0, -1)$ are also excluded. This structure is called a $1/2$ -flow-pair.

In the first part of this thesis, we develop a generator of cubic multipoles with a better performance comparing to other known generators, since the multipoles are of a great relevance in the nowhere-zero flow area.

Then, we explore the properties of $1/2$ -flow-pairs. We prove their existence on snarks with oddness 2, and we also show that one can construct a rich nowhere-zero 7-flow from them. Then we find a lower bound on two-dimensional Chebyshev flow number depending on the order of a snark. We also use a generalised version of the $1/2$ -flow-pair to investigate the Chebyshev flow number, to show that the two-dimensional Chebyshev flow number of snarks can be arbitrarily small, and to construct a Chebyshev flow on generalised Blanuša snarks.

Keywords: graph theory, flows, oddness, rich flows, lower bound, generalised Blanuša snarks, generator, multipole

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Introduction

A *flow* on a graph is an assignment of flow values to the darts of the graph. For convenience, each edge is replaced by two darts, and when we use a flow value on some edge, we can choose the dart in the desired direction. The flow must satisfy two conditions – flow values on opposite darts are additive inverses and the sum of flow values surrounding a vertex is always zero. When the flow value of some edge is zero, then the given flow is de facto a flow on the graph after deleting that edge. Hence, we usually require the flow to be *nowhere-zero*. Moreover, we can restrict the range of flow values permitted. A *nowhere-zero k -flow* can use only integral values between 1 and $k - 1$ and their additive inverses. The smallest number k such that a given graph admits a k -flow is called a flow number.

Research on integral nowhere-zero flows started in the early 1950s by Tutte [30]. This introductory paper was followed by another paper of Tutte, in which he described the relationship between the nowhere-zero flows and vertex colourings of graphs and conjectured that each bridgeless graph admits a nowhere-zero 5-flow [34].

The main problem in this area is finding the best upper bound on the flow number. Since there are graphs that do not admit a 4-flow, the bound cannot be better than 5. One of the first proved bounds was the one of Jaeger [13], when he proved that every bridgeless graph has a nowhere-zero 8-flow. Seymour [29] improved this result, showing that each bridgeless graph allows a nowhere-zero 6-flow. Note that the gap between the known and conjectured results was narrowed to only two possibilities – either there exists a nowhere-zero 5-flow on any bridgeless graph, or there is a counterexample that necessarily requires a flow value 5 or -5 .

There are several properties proved about the minimal counterexample to the 5-flow conjecture (if there exists any) – it must be cubic, not 3-edge-colourable, cyclically 6-edge-connected [16], of girth at least 12 [17] and oddness at least 6 [24]. However, it is not even known whether a graph with such properties exists.

The possible counterexample to the flow problem needs to be very large. The smallest cubic graph of girth at least 12, the Benson graph, has 126 vertices [1]. Moreover, this graph is colourable. Hence, it is not expected that we would be able to find the counterexample by an exhaustive generation of all snarks, since the currently fastest generators `tetration` and `minisnark` [5] are able to generate snarks up to 40 vertices.

However, also the local structure of the graph may be interesting in the investigation of the problem, which motivates the generation of cubic multipoles. Although there exist some generators (mainly the **pregraphs** [6] and the unpublished **multigraph**), they are not fast enough and the **pregraphs** has no option to prescribe minimal girth (which is obviously the crucial part). Therefore, we develop a generator **polehunter**, for now able only to generate multipoles of minimal girth up to 5, which is faster than the currently known generators.

In the meantime, people tried to relax the constraints of the flow problem. One of the first ideas was to use fractional flow values [11]. There were also attempts to use real flow values, but Steffen showed that this relaxation does not help [31].

Another relaxation assigns multidimensional flow values. This idea was brought to light by Jain [15], who conjectured that in three dimensions, Euclidean unit vectors are sufficient to construct a flow on any bridgeless graph. Thomassen [33] proved this in seven dimensions. However, these papers focused only on the unit vectors and did not generalise the concept to the graphs, which do not admit a flow using only unit vectors.

Since in three dimensions, it is already conjectured that all graphs admit the unit vector flow, the first paper concerning multidimensional flows not using only unit vectors focused on the two-dimensional case [23]. Considering a 2-dimensional r -flow, the bounding criterion on the permitted flow values is their Euclidean norm, which must be at least 1 and at most $r - 1$. In this setting, the flow number is often irrational. That indicates that the choice of the norm is not ideal.

Therefore, in recent preprints, the idea of Manhattan and Chebyshev flows appeared [10, 18]. With respect to these norms, the flow number is always rational. The proved and the conjectured upper bounds are 3 and $5/2$, respectively. The unit vectors are proved to be sufficient also in seven dimensions w.r.t. the Manhattan norm and in three dimensions w.r.t. the Chebyshev norm. Again, the 2-dimensional problem is interesting for snarks, since for and only for colourable cubic graphs, a flow composed of unit vectors always exists. The research also led to a pair of flows, where the first one uses values from $\{0, \pm 1\}$, the other values $\{0, \pm 1, \pm 2, \pm 3\}$, but moreover, $(0, 0)$, $(0, 1)$ and $(0, -1)$ are forbidden. This so-called *flow-pair* implies the existence of 5-flow and two-dimensional Chebyshev $5/2$ -flow.

In this work, we show other interesting relations of this flow-pair. We prove that the flow-pair exists on snarks of oddness 2. For graphs with a flow-pair, we are also able to construct a nowhere-zero 7-flow not using the same element around a vertex twice (called a *rich* flow). Thus, in total, graphs of oddness 2 admit a rich nowhere-zero 7-flow.

As mentioned above, snarks are exactly those cubic graphs, for which flow number is greater than two. One interesting question is, how small it could be. We managed to

prove a lower bound dependent on the order of a graph, already submitted to a journal [10], and similar to a one-dimensional lower bound on flow number [21]. We also use a generalised version of the flow-pair to construct an infinite class of snarks with flow numbers, that can be arbitrarily small.

Chapter 1

Preliminaries

Unless stated otherwise, we consider unoriented simple connected graphs $G = (V, E)$. Then we can replace each unoriented edge uv with two oriented edges $\overrightarrow{uv}$, $\overrightarrow{vu}$, called *darts*. By a *dart-set* $D(S)$ of $S \subseteq V$ we mean the set of all darts with initial vertex in the set S . Finally, we define $D(G)$ as $D(V)$. A *subgraph* of a graph is a graph $G' = (V', E')$ with $V' \subseteq V$ and $E' \subseteq E$. The maximal degree of a graph is denoted by $\Delta(G)$. A graph is called *r-regular*, if each of its vertices is of degree r . A 3-regular graph is also called *cubic*. Due to an easy parity argument, each cubic graph has even order. A *k-factor* is a k -regular subgraph of a graph on the same set of vertices. A *dangling edge* is an edge with only one endpoint in a vertex of the graph. A *k-pole* is a graph containing k dangling edges. A *complete graph on n vertices* K_n consists of all possible $\binom{n}{2}$ edges.

For a graph G and its edge e , a graph $G - e$ is obtained by removing the edge e without removing its endvertices. For a cubic G , we can also suppress these two endvertices (which are of degree 2) to obtain $G \sim e$. These operations are called *edge deletion* and *edge reduction*, respectively.

A *girth* is the length of the shortest cycle in a graph. Observe that a 2-factor is a set of cycles covering all vertices exactly once. The smallest possible number of cycles on odd vertices in a 2-factor of a cubic graph is called the *oddness*. Since cubic graphs have even order, the oddness must be even.

A *bridge* is an edge whose removal disconnects the graph. Similarly, an *edge-cut* is a set of edges disconnecting the graph after its removal. An *edge-connectivity* is the minimal number of edges in the edge-cut of the graph, and a *k-edge-connected* graph is a graph with the edge-connectivity not smaller than k . A *2-edge-connected graph* can also be called *bridgeless*. For a *cyclic edge-connectivity*, we also require the components to contain a cycle – so they cannot be trees. An *uv-edge-separator* is an edge-cut, for which u and v are in different components. Menger's theorem [26] relates the size of the smallest *uv-edge-separator* to the largest number of disjoint *uv*-paths. We need its

oriented version, so we state it below as Theorem 1.1.

Theorem 1.1 (Menger's theorem for oriented graphs). [9, p. 149] *Let u, v be the vertices of an oriented multigraph G . Then there exists k edge-disjoint oriented uv -paths if and only if for any set $S \subseteq V(G) \setminus v$ containing u , there are at least k oriented edges from S to $V(G) \setminus S$.*

An *edge-colouring* of a graph is an assignment from a set of edges E to a set of colours C . The edge-colouring is *proper* if no adjacent edges share a colour. The minimal k , such that a graph has a proper edge-colouring using k colours, is a *chromatic index* $\chi'(G)$. The chromatic index is bounded by $\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$ [35], and so can attain only two values. Therefore, graphs with $\chi'(G) = \Delta(G)$ are called *class one*, or *colourable* graphs, and the others are called *class two*, or *uncolourable* graphs. A snark is an uncolourable cubic graph which is cyclically 4-edge-connected and has girth at least 4. Since cubic graphs having oddness 0 are colourable, snarks should have oddness at least 2. A snark is *weak* if it has an edge e such that $G \sim e$ is colourable. Otherwise, the snark is called *strong*.

An *isomorphism* of graphs $G = (V, E), G' = (V', E')$ is a mapping $\sigma: V \rightarrow V'$ preserving adjacency (i.e. uv is an edge in G if and only if $\sigma(u)\sigma(v)$ is an edge in G'). Graphs are called *isomorphic* if there exists an isomorphism mapping one to another. An *automorphism* of a graph is an isomorphism from this graph to itself. Vertices u_1, u_2 are in the same *vertex orbit* of a graph, if there exists an automorphism with $\sigma(u_1) = u_2$. Similarly, edges u_1v_1, u_2v_2 are in the same *edge orbit* of a graph, if there exists an automorphism with $\sigma(u_1v_1) = u_2v_2$. Graphs with single vertex orbit are called *vertex-transitive*, while the ones with single edge orbit are called *edge-transitive*.

The *Petersen graph* (in Figure 1.1) is the smallest snark. This graph is vertex-transitive and also edge-transitive.

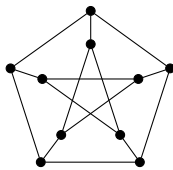
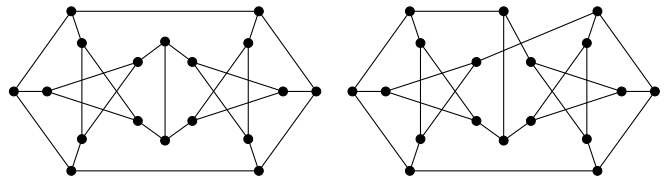


Figure 1.1: Petersen graph



(a) first (b) second

Figure 1.2: Blanuša snarks

Blanuša snarks (in Figure 1.2) are the smallest snarks except the Petersen graph. They are a part of two infinite classes of snarks, called *generalised Blanuša snarks*, constructed from three types of 4-poles depicted in Figure 1.3. The n -th *generalised Blanuša snark of type I* $B_n^{(1)}$ is constructed from n copies of the 4-pole B and one copy of the 4-pole I by identifying consecutive a with $\bar{a}$ and b with $\bar{b}$. Similarly, we obtain the n -th *generalised Blanuša snark of type II* $B_n^{(2)}$ from $n - 1$ copies of the 4-pole B and

one copy of the 4-pole X . For both classes, the first graph in the class is the Petersen graph, and the second one is the corresponding Blanuša snark.

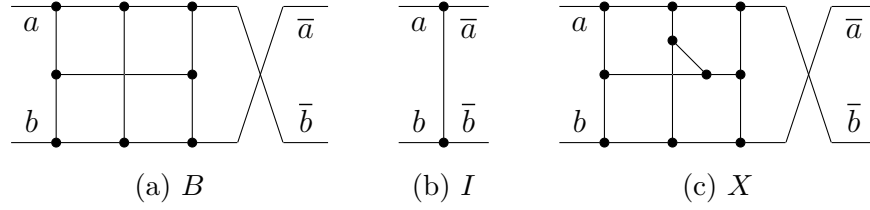


Figure 1.3: Three 4-poles used to construct generalised Blanuša snarks

The *cage problem* is a problem of finding the smallest k -regular graphs with girth at least g . Such graphs are called (k, g) -*cages*. The lower bound on the order of a (k, g) -cage follows from the fact, that up to a distance $\lfloor \frac{g-1}{2} \rfloor$, the graph locally is a tree. This tree is called a *Moore tree*, the corresponding bound is called a *Moore bound* and graphs attaining this bound are called *Moore graphs*.

Chapter 2

Cubic graphs generation

In this chapter, we introduce methods for generating cubic graphs with a desired minimal girth. A program called `minibaum` [2] is based on the idea of the Moore tree, while `snarkhunter` [3, 4, 7] generates the graphs from a few elementary ones by adding edges.

2.1 Minibaum

The generator `minibaum` [2] is one of the first large-scale generators of regular graphs, invented already in the previous century. Given the number of vertices, the regularity, and the minimal girth, it generates all feasible graphs. The main idea of this algorithm is a representation of a labeled graph as its adjacency matrix, which is the lexicographically largest one out of all isomorphic options. Such a representation is called *maximal*.

The generation starts with a maximal Moore tree corresponding to the prescribed regularity and girth. Then, the authors prove a lemma [2, p. 143] that in the maximal representation, the girth cycle is the cycle with the lexicographically smallest vertex labels. Therefore, after constructing the first girth cycle, graphs obtained from this graph are either of the prescribed girth or are not in their maximal representation.

2.2 Snarkhunter

One of the most efficient generators of cubic graphs, as for now, is `snarkhunter` [3, 4, 7]. Its main idea follows from the fact, that by the edge reduction $G \sim e$ the graph remains in the class of cubic graphs, but its order decreases. However, the resulting graph may be disconnected, have multiple edges, or loops.

Definition. [4, p. 70] Let G be a cubic graph and e its edge. We say that e is *reducible* if the graph $G \sim e$ is simple and connected.

The natural question is, how can we easily determine an edge that is not reducible. Let us check it case by case.

- $G \sim e$ is disconnected. Then, since G is connected, the removal of the edge e disconnects the graph G , and so e must be a bridge in G . Conversely, if e is a bridge in G , then the graph $G \sim e$ is clearly disconnected.
- $G \sim e$ has a loop e' in a vertex x . Then, in G , at least one endvertex of $e = uv$ lies on e' . When exactly one endvertex of e lies on e' (WLOG u), then there are multiple edges between u and x . Otherwise, there are multiple edges between u and v . Hence, the original graph G cannot be simple, a contradiction.
- $G \sim e$ has multiple edges e_1, e_2 connecting x and y . Then, in G , at least one endvertex of $e = uv$ lies either on e_1 or on e_2 .
 - One endvertex of e (WLOG u) lies on one of the multiple edges (WLOG e_1). Then, x, y , and u form a triangle in G . Conversely, if e is an edge with exactly one vertex in a triangle in G , the graph $G \sim e$ obviously contains multiple edges.
 - Both endvertices of e lie on the multiple edges (WLOG u on e_1 , v on e_2). But then, x, u, y, v form a square in G and $e = uv$ is its diagonal. Conversely, if e is a diagonal of a square in G , the resulting graph $G \sim e$ contains multiple edges.

Now, we are ready to define “elementary” graphs. Following the convention from number theory, they are called *prime* graphs.

Definition. [4, p. 70] A cubic graph G is called *prime* if it has no reducible edge.

Hence, from those prime graphs, we can inductively construct any cubic graph by the reverse of the edge reduction – we subdivide two (possibly adjacent) edges and join the new vertices by an edge. We call this an *edge extension*. The only remaining issue is generating prime poles.

We introduce two cubic multipoles relevant for the characterisation of prime graphs, depicted in Figure 2.1. A *diamond* is a 2-pole D arising from K_4 by splitting its edge into two dangling edges. Since K_4 is vertex-transitive, the definition of diamond does not depend on the chosen edge, and therefore is uniquely determined. On the other hand, by subdividing any edge of K_4 with a vertex and connecting this vertex to a new vertex, we get a *lollipop* 1-pole L . Observe that the lollipop also contains a diamond.

As mentioned above, now we are able to characterise prime graphs.

Lemma 2.1. [4, p. 70] *A cubic graph is prime if and only if each of its edges is either a bridge or contained in a diamond.*

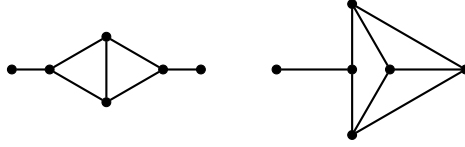


Figure 2.1: Diamond (left) and lollipop (right) multipoles

However, the prime graphs can also be characterised in an inductive way, which allows us to generate them from several base cases.

Lemma 2.2. [4, p. 70] *Each prime graph can be obtained from a K_4 by applying a finite number of extensions (a), (b), and (c) depicted in Figure 2.2. Moreover, the extensions can be applied in a way such that all intermediate graphs are also prime.*

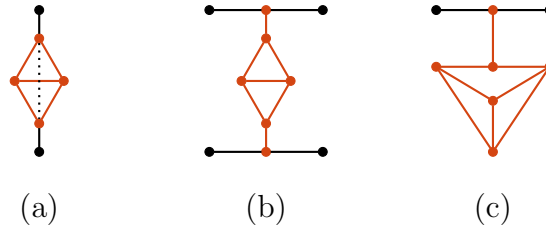


Figure 2.2: Prime graph extensions

The number of prime graphs is rapidly smaller than the number of all cubic graphs – for illustration, there are 753 prime graphs on at most 38 vertices, although there are cca. 300 quadrillions of cubic graphs on 38 vertices.

We describe the overall idea of the recursive algorithm for the edge extension, since the prime graph extensions work analogously. Having a graph G , we generate all possible pairs of edges (called *edgepairs*), where the edge extension can be applied. Observe that when applying the edge extension to edgepairs (e_1, e_2) and (e'_1, e'_2) , where e_1, e'_1 and e_2, e'_2 are in the same edge orbits, the resulting graphs are isomorphic. Therefore, for each combination of edge orbits, the extension is applied only on one of the edgepairs.

The other way to get two isomorphic graphs of order n during the generation is to obtain them by extensions from two non-isomorphic graphs of order $n - 2$. To decide which of those extensions should be accepted, a canonical labeling of vertices provided by `nauty` [25] is used.

However, determining a canonical labeling of vertices is a computationally hard problem (since otherwise, graph isomorphism would not be in NP), so there is an aim to avoid computing canonical labels as often as possible. This can be done by computing some local invariants for the edges, because when two edges have a different invariant, they can not be in the same edge orbit. The invariants used by the authors of `snarkhunter` with the corresponding criteria are in Figure 2.3.

- x_0 an indicator whether the edge is in a 4-cycle (largest),
- x_1 the number of vertices in distance at most 2 from the edge (smallest),
- x_2 the number of 4-cycles containing the edge (largest),
- x_3 the number of vertices in distance at most 3 from the edge (least frequent),
- x_4 the number of vertices in distance at most 4 from the edge (smallest),
- x_5 the sum of x_1 and x_3 through edges adjacent to the edge (smallest), and
- $x_{6,7}$ the lexicographically largest **nauty** vertex labels $x_6 > x_7$ of the edge in the same edge orbit (largest)

Figure 2.3: Edge extension invariants

This idea distinguishes edgepairs considering the same reduction operation. But, besides edge reductions, the algorithm uses also prime graph reductions and so-called *triangle reductions*. We need to set priorities of these reductions.

Note that the edge extension applied to a pair of adjacent edges results in a triangle. However, this extension is equivalent to substituting the common vertex of edges with a triangle. Moreover, it can be shown that almost all triangles can be reduced simultaneously. The irreducible ones are the triangles contained in some diamond. Then, there are pairs of reducible triangles, which cannot be reduced simultaneously – those contained in an *extended diamond* depicted in Figure 2.4.

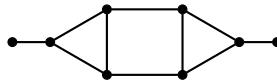


Figure 2.4: Extended diamond

However, reducing any of the triangles in Figure 2.4 results in a diamond, so it is not dependent on its choice. Hence, the triangle operation reduces as many triangles as possible, and then the result is deterministic, so we do not need to find the canonical labeling. Therefore, if there are reducible triangles, authors prefer the triangle reduction over the edge reduction. Moreover, for the general case, we can assume that edges in the edgepair are disjoint.

Back to the priority of reductions. The triangle reduction has the highest priority, and is followed by the edge reduction. A reduction of a prime graph is accepted only if it results in another prime graph. Finally, the order of prime graph reductions is (c) > (b) > (a) – so if a prime graph contains a lollipop, the reduction must reduce some

lollipop.

Another time consuming part of the algorithm is generating feasible edge-pairs. The authors divide the generation into multiple special cases, in which they can reduce the number of possible edgepairs from quadratic to either linear or constant. For each of these cases, they implement a tailored method. For an illustration, in Figure 2.5, we give a structure of 7 methods called when generating edgepairs.

Depending on the number of reducible triangles in G , the corresponding method is called:

- *two triangles* – only 6 edgepairs destroying both of them,
- *one triangle* – edgepairs consist of one edge from the triangle and one outside,
- *no triangles* – edgepairs contained in a 4-cycle are generated, and then, depending on the number of squares in G , the corresponding method is called:
 - *two adjacent squares* – only 1 valid edgepair,
 - *two non-adjacent squares* – only 16 edgepairs destroying both of them,
 - *one square* – edgepairs consist of one edge from the square and one outside,
 - *no squares* – all edgepairs must be generated.

Figure 2.5: Structure of methods generating edgepairs

2.3 Extension of multiple edges at once

Note that in the backtracking tree of the algorithm, the main part of the generation is taken by the last levels, caused by the rapid growth of the graphs count. However, the authors of **snarkhunter** show that having enough girth, the edge reduction of more edges can be done locally for each cubic graph. Therefore, the number of possible candidates for extensions is smaller, and the algorithm runs faster [3].

A *tripod* subgraph is a 3-pole $K_{1,3}$, and an $\mathcal{H}$ subgraph is the only acyclic cubic 4-pole. Observe that a tripod and an $\mathcal{H}$ extensions (see Figure 2.6) simulate consecutive extensions of two and three edges, respectively. Hence, their reductions simulate consecutive edge reductions. These subgraphs are called *reducible* in a graph G if the reduced graph remains simple and connected.

Since each graph has a vertex that is not a cutvertex, there is a reduction of the tripod not disconnecting the original graph. Moreover, since the girth is large enough, the reduction produces no multiple edges.

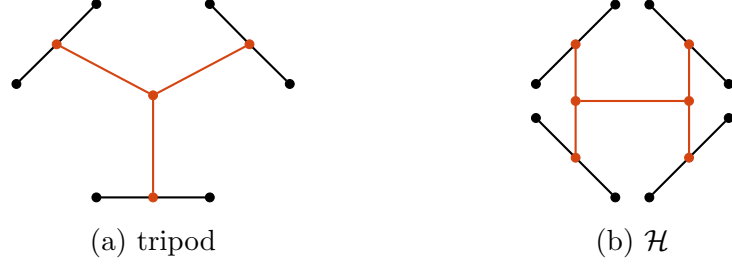


Figure 2.6: Illustration of multiple extensions

Theorem 2.3. [3, p. 260] *Each connected cubic graph with girth $g \geq 5$ has a reducible tripod. Moreover, the girth of the reduced graph is at least $g - 1$, and the number of $(g - 1)$ -cycles is at most 3.*

A similar result can be proved for the $\mathcal{H}$ reduction. Here, the authors define an edge to be *good* when, after removing both of its endvertices, the graph remains connected. Observe that this is a necessary and a sufficient condition for the graph after the $\mathcal{H}$ reduction to remain connected.

Lemma 2.4. [3, p. 264] *Each connected cubic graph has a good edge.*

However, with an additional assumption of a minimal girth, the reduction of $\mathcal{H}$ cannot produce multiple edges.

Theorem 2.5. [3, p. 264] *Each connected cubic graph with girth $g \geq 5$ has a reducible $\mathcal{H}$. Moreover, the girth of the reduced graph is at least $g - 2$ for $g \leq 6$ and $g - 1$ otherwise.*

From the computational results, the tripod reduction does not help much for girth 5, so the actual implementation of `snarkhunter` uses edge reductions for girths up to 5 and $\mathcal{H}$ reductions for girths 6 and 7.

When deciding whether the reduction is canonical, the algorithm uses invariants similar to those in Figure 2.3. We do not describe them, since we see no benefit of it.

2.4 Other features of snarkhunter

The `snarkhunter` allows a computation distribution into multiple machines. This is done by setting a *splitlevel*. When the algorithm generates a graph having the order of this splitlevel, a counter of graphs on the splitlevel is incremented. Then, the extensions are applied to this graph only if the value of this counter has a prescribed remainder modulo a given number.

The generation of snarks before `snarkhunter` was only possible by generating all cubic graphs of prescribed order, and then filtering them. However, the edge extension

is closely related to the edge-colouring of the graph, and so many colourable graphs can be discarded by a simple look-ahead [7].

There is also an option to generate only graphs with a greater (cyclic) edge-connectivity.

Chapter 3

Flows

In this chapter, we aim to introduce all concepts and current knowledge about flows and related graph theory terms. Mainly, we explain *Chebyshev nowhere-zero flows*, which have arisen in the recent preprint coauthored by the author [10]. Moreover, we present a *flow-pair* – a sufficient condition for a graph to have some nowhere-zero flow and some Chebyshev flow, also stated in that paper. We also establish a notion of *rich flows* not allowing same flow values on darts around a vertex.

3.1 Group flows

Intuitively, flows represent the dynamics of a fluid in some network. Consider that x gallons of water from a vertex u flow to a vertex v . When we ask, how many gallons of water from a vertex v flow to a vertex u , we should say $-x$. Hence, when working with flows, we naturally need inverses. Moreover, if the system is stable, the amount of water that flows in should be the same as the amount of water that drains out. This can be reformulated as “(possibly negative) amounts of water flowing into a vertex sum up to zero”. This motivates the need for a neutral element. So, we end up with a group. Therefore, the original definition of nowhere-zero flows uses a group-theory setting with additive notation.

Definition. Let $G = (V, E)$ be a graph, $\mathcal{A}$ be an abelian group, and $F \subseteq \mathcal{A}$ be the set of permitted flow values closed under inverse. An $(\mathcal{A}, F)$ -flow on G is an assignment $\varphi: D(G) \rightarrow F$ of flow values to the darts satisfying $\varphi(\overrightarrow{uv}) = -\varphi(\overrightarrow{vu})$ for any dart $\overrightarrow{uv} \in D(V)$ and

$$\sum_{d \in D(\{v\})} \varphi(d) = 0 \tag{3.1}$$

for each vertex v of G . The equality (3.1) is often referred to as the *flow conservation constraint*. If $F = \mathcal{A}$, we use an abbreviation $\mathcal{A}$ -flow, and if $F = \mathcal{A} \setminus 0$, we use the term *nowhere-zero $\mathcal{A}$ -flow* (or abbreviated *NZ $\mathcal{A}$ -flow*).

Observe that the nowhere-zero condition is crucial in avoiding trivial flows, since otherwise, we are able to set all flow values to zero, and it is a valid flow. The flow conservation constraint (3.1) is generalisable to any subset of vertices. Hence, having a graph G with a bridge e , the interpretation of the flow conservation constraint for one of the components of $G - e$ results in the fact, that G admits no NZ flow. Therefore, nowhere-zero flows are researched only on bridgeless graphs. However, by an interesting argument, the number of NZ $\mathcal{A}$ -flows of some graph depends only on the order of the group.

Theorem 3.1. [34, p. 82] *Let $\mathcal{A}_1, \mathcal{A}_2$ be two finite abelian groups of the same order and G be a graph. Then G admits an NZ $\mathcal{A}_1$ -flow if and only if it admits an NZ $\mathcal{A}_2$ -flow.*

3.2 Integral flows

Results in the previous section motivated, that the specific group is not relevant, relevant is only its order. We can use this to get out of the group setting and replace its elements by integers.

Definition. Let $k \geq 2$ be a positive integer. A k -flow is a $(\mathbb{Z}, \{0, \pm 1, \pm 2, \dots, \pm(k-1)\})$ -flow. Moreover, a *nowhere-zero k -flow* (or *NZ k -flow*) is a $(\mathbb{Z}, \{\pm 1, \pm 2, \dots, \pm(k-1)\})$ -flow.

Since integers are well-ordered, we also establish a notation $\varphi(e)$ for $e = uv \in E$, where $\varphi(e) = \max\{\varphi(\overrightarrow{uv}), \varphi(\overrightarrow{vu})\}$. Hence, we can also take a *positive* flow – for each edge, we take a dart with the positive flow value. Now, to see that this newly defined notion truly replaces the problem in the group theory setting, we need the following theorem.

Theorem 3.2. [13, p. 207] *For each $\mathbb{Z}_k$ -flow φ there exists a k -flow φ' satisfying $\varphi'(d) \equiv \varphi(d) \pmod{k}$.*

Remark. Since the preceding theorem preserves modulo, it also preserves zeroes, and hence the same holds for NZ flows.

It is not hard to see that a graph with an NZ k -flow also admits an NZ K -flow for any $K > k$. Hence, it makes sense to seek the smallest k such that a graph admits an NZ k -flow.

Definition. Let G be a bridgeless graph. A *flow number* of G is

$$\Phi(G) := \min\{k \mid \exists \text{ NZ } k\text{-flow on } G\}.$$

However, it turns out that this rate of complexity achieves only a few values.

Theorem 3.3 (6-flow theorem). [29, p. 133] *There exists an NZ 6-flow on any bridgeless graph.*

Seymour constructed this NZ 6-flow from two simpler flows from the fact, that if $a \in \{0, \pm 1\}$, $b \in \{0, \pm 1, \pm 2\}$ and not both a, b are zero, then $3a + b \in \{\pm 1, \pm 2, \pm 3, \pm 4, \pm 5\}$.

Lemma 3.4. [29, p. 132] *For any bridgeless graph, there exist a $(\mathbb{Z} \times \mathbb{Z}, (\{0, \pm 1\} \times \{0, \pm 1, \pm 2\}) \setminus \{(0, 0)\})$ -flow.*

On the other hand, there is no known graph with $\Phi(G) = 6$, all known graphs have $\Phi(G) \leq 5$. Therefore, the following conjecture about the NZ 5-flow by Tutte still holds unresolved.

Conjecture 3.5 (5-flow conjecture). [34, p. 83] *There exists an NZ 5-flow on any bridgeless graph.*

As for many problems in graph theory, it is sufficient to explore the problem only on a smaller subset of graphs. There exists a folklore reduction of nowhere-zero flows to cubic graphs.

Proposition 3.6. *There exists an NZ k -flow on any bridgeless graph if and only if there exists an NZ k -flow on any cubic bridgeless graph.*

So if there is a counterexample to Tutte's 5-flow conjecture, there is also a cubic counterexample. Moreover, in terms of finding such counterexample, we can also avoid colourable ones and limit research to snarks, as a corollary of the next proposition.

Proposition 3.7. [8, pp. 160, 161] *A cubic graph has an NZ 4-flow if and only if it is 3-edge-colourable.*

There are more constraints regarding the smallest counterexample of Conjecture 3.5. One of the first results concerns snarks of small oddness.

Theorem 3.8. [14, p. 89] *Let G be a snark of oddness 2. Then, $\Phi(G) \leq 5$.*

As a consequence, since each weak snark has oddness 2 (which is considered to be folklore), this also covers all weak snarks. Hence, any counterexample to 5-flow conjecture must be a strong snark of oddness at least 4. Moreover, we know some properties of the potential smallest counterexample to the 5-flow conjecture.

Theorem 3.9. [16, p. 142] *If the 5-flow conjecture is false, the smallest counterexample is a cyclically 6-edge-connected snark.*

Theorem 3.10. [24, p. 364] *If the 5-flow conjecture is false, the smallest counterexample has oddness at least 6.*

Theorem 3.11. [17, p. 15] *If the 5-flow conjecture is false, the smallest counterexample has girth at least 12.*

3.3 Chebyshev flows

There were some attempts to generalise nowhere-zero flows to real numbers and more dimensions. By generalising integer nowhere-zero flows to real numbers, researchers found out that restricting only zero flow value is not sufficient, since any flow consisting of non-zero flow values can be made “smaller” and then, the notion of flow number disappears. Therefore, we need to require an interval of forbidden flow values, and hence, one-dimensional real nowhere-zero flows are allowed to use flow values from $[1-r, r-1] \setminus (-1, 1)$ [11]. The generalisation of real flows to more dimensions [23] picks only vectors with the Euclidean norm from the mentioned interval $[1, r-1]$. It works, but it did not have such nice properties, as we would expect. However, changing a norm turned out to be a good idea [10], so we will consider only Chebyshev multidimensional nowhere-zero flows.

Definition. A d -dimensional open and closed balls of radii r with respect to Chebyshev norm are

$$B_d^\infty(r) = \{x \in \mathbb{R}^d \mid \|x\|_\infty < r\}, \text{ and} \\ \overline{B}_d^\infty(r) = \{x \in \mathbb{R}^d \mid \|x\|_\infty \leq r\},$$

respectively.

Definition. Let d be a positive integer and $r \geq 2$ be a real constant. A d -dimensional Chebyshev nowhere-zero r -flow (or dD ChNZ r -flow) is a $(\mathbb{R}^d, \overline{B}_d^\infty(r-1) \setminus B_d^\infty(1))$ -flow.

Now, for $e = uv$, the notation $\varphi(e)$ denotes the lexicographically larger one from $\varphi(\overrightarrow{uv}), \varphi(\overrightarrow{vu})$. The flow number can be defined in a similar way to integral nowhere-zero flows. The only problem is that if we take a set $\{r \mid \exists dD \text{ ChNZ } r\text{-flow on } G\}$, it may not have a minimum, so a formally correct definition would use infimum instead of minimum. However, the infimum of the flow number candidates is always feasible [23, p. 101], so we can define the flow number straightforwardly with the minimum.

Definition. Let G be a bridgeless graph and d be a positive integer. A d -dimensional Chebyshev flow number of G is

$$\Phi_d^\infty(G) := \min\{r \mid \exists dD \text{ ChNZ } r\text{-flow on } G\}.$$

One special case worth mentioning is a *circular NZ flow* – a 1D ChNZ flow. In one dimension, it does not matter, which norm is used to constrain permitted flow values, so we can omit it from the naming. Similarly, Φ_1^∞ can be replaced by Φ_1 . As we show in the following proposition, there is a relation between circular flow number and integral flow number of a graph – namely $\Phi(G) = \lceil \Phi_1(G) \rceil$.

It has been proved [10, p. 4] that the Chebyshev flow number is always rational. Moreover, the authors found a normal form of Chebyshev flows, as stated in the following proposition.

Proposition 3.12. [10, p. 6] *Let G be a graph with a ChNZ p/q -flow φ for some positive integers p, q . Then, there exists a dD ChNZ p/q -flow such that every coordinate of every flow value is one of the two nearest integral multiples of $1/q$ to the corresponding coordinate in φ .*

Now we recall Seymour's flows from Lemma 3.4 – a 2-flow and a 3-flow with a property that no edge has both flow values equal to zero. By simply using the value of the 2-flow as the first coordinate and the value of the 3-flow as the second coordinate, we get a Chebyshev nowhere-zero flow. Moreover, this flow gives us a nice bound on the Chebyshev flow number.

Proposition 3.13. [10, p. 8] *Each bridgeless graph admits a 2D ChNZ 3-flow.*

Again, no graph with $\Phi_2^\infty(G) = 3$ is known. Furthermore, there is no known graph with $\Phi_2^\infty(G) > 5/2$. Therefore, the next conjecture is interesting to examine.

Conjecture 3.14. [10, p. 22] *Each bridgeless graph admits a 2D ChNZ $5/2$ -flow.*

Similarly to integral flows, the problem can be reduced to cubic graphs and then to snarks, since colourable cubic graphs have a small flow number.

Theorem 3.15. [10, p. 14] *A cubic graph has a 2D ChNZ 2-flow if and only if it is 3-edge-colourable.*

3.4 Flow-pairs

As an NZ 6-flow and a 2D ChNZ 3-flow can be constructed from two simpler flows, something similar can be made for smaller flow numbers with two other flows [10, p. 19].

Definition. Let $p \leq q$ be positive integers. A p/q -flow-pair of a graph G is a $(\mathbb{Z} \times \mathbb{Z}, F)$ -flow with

$$F = \{(0, f) \mid q \leq |f| \leq p + q\} \cup \{(1, f) \mid |f| \leq p + q\}.$$

Remark. The structure in Lemma 3.4 is a $1/1$ -flow-pair.

One possible ambiguity of this definition is that since p/q is written as a fraction, the $2/6$ -flow-pair can be mixed up with the $1/3$ -flow-pair. However, their existence is equivalent (by a similar argument to Proposition 3.12), so we can WLOG assume that p, q are relatively prime.

Lemma 3.16. [10, p. 19] *A bridgeless graph G with a dp/dq -flow-pair also has a p/q -flow-pair and vice versa.*

We mentioned that the p/q -flow-pair is a sufficient condition for the existence of some nowhere-zero flows. Those flows are somehow related to the value p/q , and that is the reason to write it as a fraction in the definition of the flow-pair.

Proposition 3.17. [10, pp. 19, 20] *Consider a bridgeless graph G with a p/q -flow-pair $\varphi = (\varphi_1, \varphi_2)$. Then, there are a 2D ChNZ $(2 + p/q)$ -flow $\varphi' = (\varphi_1, \varphi_2/q)$ and a circular NZ $(4 + 2p/q)$ -flow $\varphi'' = (2 + p/q)\varphi_1 + \varphi_2/q$ on G .*

Note that if there is a graph, which is a counterexample to Tutte's 5-flow conjecture, it should not have the $1/2$ -flow-pair. Hence, it would be interesting to classify graphs with $1/2$ -flow-pair. By a computer, it was checked that this flow-pair exists for snarks up to 34 vertices, so it looks like the $1/2$ -flow-pair exists always.

Conjecture 3.18. [10, p. 22] *Each bridgeless graph has a $1/2$ -flow-pair.*

In general, edges with a non-zero value of the first coordinate form a set of cycles in G . In cubic graphs, these cycles should be clearly vertex-disjoint. Moreover, if considering other than the trivial $1/1$ -flow-pair, these cycles together cover all vertices.

Lemma 3.19. [10, p. 21] *Let $p < q$ be positive integers. Consider a cubic graph G with a p/q -flow-pair $\varphi = (\varphi_1, \varphi_2)$. Then the set of edges e with a property $\varphi_1(e) \neq 0$ is a 2-factor of G .*

3.5 Rich flows

Recently, in the research of flows, there was an idea to forbid two darts surrounding a vertex to have flow values with the same absolute value [19, 28]. The motivation is that if we have two darts from the same vertex with $\varphi(d_1) + \varphi(d_2) = 0$ (a so-called *confluent pair*), we can split the considered vertex of the degree d to vertices of degrees 2 and $d - 2$ without influencing flow properties of a graph. On the other hand, a so-called *contrafluent pair*, where $\varphi(d_1) = \varphi(d_2)$, requires that the set of flow values contains some element and its double. However, we cannot find such a pair in a p/q -flow-pair with $p < q$.

Definition. An NZ k -flow φ is called *rich* if for each pair of adjacent darts d_1, d_2 it holds that $|\varphi(d_1)| \neq |\varphi(d_2)|$.

We already know that a graph admits a nowhere-zero flow if and only if it is bridgeless. In the theory of rich flows, there is one more obstruction – a 2-edge-cut with the common vertex. It can be shown that these two conditions are also sufficient for a graph to have the rich nowhere-zero flow. We call such graphs *rich flow admissible*.

Definition. Let G be a rich flow admissible graph. A *rich flow number* of G is

$$R(G) := \min\{k \mid \exists \text{ rich NZ } k\text{-flow on } G\}.$$

Unlike group, integral, circular, and multidimensional flows, rich flows and bounding the rich flow number cannot be reduced to the class of cubic graphs. The simple reason is that we clearly need at least $\Delta(G)$ different flow values and therefore $R(G) \geq \Delta(G) + 1$. On the other hand, there is a linear upper bound depending only on $\Delta(G)$ [19].

However, if we limit our view only to the cubic graphs, we get similar results as for integral nowhere-zero flows. Again, it suffices to examine the snarks.

Proposition 3.20. [28, p. 31] *A cubic graph has a rich NZ 5-flow if and only if it is 3-edge-colourable.*

And from the computational results, the gap between the colourable and uncolourable cubic graphs is again equal to 1. However, with the proven results, we are still far from the desired value 6.

Conjecture 3.21. [28, p. 33] *There exists a rich NZ 6-flow on any bridgeless cubic graph.*

Proposition 3.22. [28, p. 33] *There exists a rich NZ 11-flow on any bridgeless cubic graph.*

For snarks that are “nearly colourable”, we have a result similar to Corollary 3.8 — we can reduce the size of the gap from 5 to 3.

Proposition 3.23. [28, p. 40] *Let G be a weak snark. Then, $R(G) \leq 9$.*

Chapter 4

Cubic multipoles generator

In this chapter, we generalise the idea of **snarkhunter** generating cubic graphs to cubic multipoles, and so describe an algorithm called **polehunter**. We also reimplement the **snarkhunter** for girths 3, 4, and 5, obtaining a significantly faster generator in comparison to **multigraph** and **pregraphs** [6]. Despite not implemented, we built a background for the generation of cubic graphs of minimal girths 6 and 7.

4.1 Existing generators

Program **multigraph** generalises the idea of **minibaum**. This algorithm generates all graphs having a given degree sequence with the given minimum girth g . Hence, the generation of cubic multipoles of a prescribed girth is a special case of this algorithm. However, there is currently no paper concerning this generator [12].

Program **pregraphs** [6] is a generator based on **snarkhunter**. It generates graphs with loops, dangling edges, and multiple edges. There are also several options for filtering graphs on edge-colourability, bipartiteness, or the existence of a 2-factor consisting only of 4-cycles. This generator does not have an option to prescribe minimal girth. Therefore, it can be used only for generating all cubic multipoles.

4.2 Polehunter

We implemented an algorithm called **polehunter** (since it is a reimplement of **snarkhunter** for multipoles). In comparison to **multigraph**, since **snarkhunter** is much faster than **minibaum**, we expect that the **polehunter** will be, too. On the other hand, **pregraphs** uses **snarkhunter** as a blackbox, and then only adjusts the resulting cubic graphs by adding dangling edges. More formally, the **pregraphs** is only the reduction of the poles generation problem to the cubic graphs generation problem managed by the unchanged **snarkhunter**. Our approach reimplements the core of the

snarkhunter. Although we do not expect **polehunter** to be significantly faster than **pregraphs** (since their ancestor is the same), we consider the possibility of a minimal girth prescription as a large advantage.

The representation of a cubic pole in **polehunter** is by a graph with vertices of degrees 1 and 3, where the edges containing a vertex of degree one are exactly the dangling ones.

First, we need to restate the definitions of reducible edges and prime graphs.

Definition. Let P be a cubic pole and e its edge. We say that e is *reducible*, if the pole $P \sim e$ is simple and connected. A cubic pole is called *prime* if it has no reducible edge.

Then the characterisation of the prime poles is very similar to the one concerning cubic graphs.

Lemma 4.1. *A cubic pole is prime if and only if each of its edges is either a bridge, a dangling edge, or contained in a diamond.*

Lemma 4.2. *Each prime pole can be obtained from K_2 or K_4 by applying a finite number of extensions (a), (b), (c), and (d) depicted in Figure 4.1. Moreover, the extensions can be applied in a way such that all intermediate poles are also prime.*

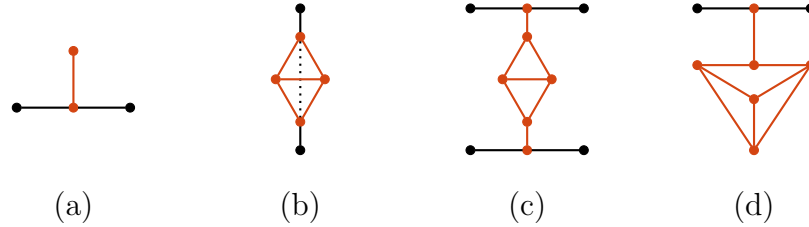


Figure 4.1: Prime poles extensions

Proof. It suffices to prove that for a given prime pole P of order greater than 2 different from K_4 , a reduction (a), (b), (c), or (d) can be applied, and the resulting pole P' remains prime.

First, assume that P has a dangling edge uv , where v has degree one. When this dangling edge is a part of a diamond, then we can reduce that diamond with (b), since the edge created by the reduction is again a dangling edge, and the other edges of the pole remain irreducible. Otherwise, the neighbours of u different from v are not adjacent (because that would create a reducible triangle), so we can reduce the dangling edge with (a). The edge created by the reduction is simple and is either a bridge, dangling, or a part of two different diamonds, and hence cannot be reduced, so the graph after reduction remains prime.

Now we are left with a prime pole having no dangling edges, and therefore we can apply Lemma 2.2, and so this pole can be constructed from a K_4 using (b), (c), and (d). $\square$

Therefore, the main load of the reimplementaion of **snarkhunter** to **polehunter** for minimal girths 3, 4, and 5 should be in the new extension operation (a).

Now we can move to the minimal girths 6 and 7. Here, the original algorithm of **snarkhunter** uses an $\mathcal{H}$ extension instead of three levels of edge extension. We need to adjust the definition of the good edge such that the central edge of $\mathcal{H}$ is not dangling.

Definition. An edge uv of a cubic pole P is called *good* if both its endvertices are of degree 3 and the graph $P - \{u, v\}$ contains at most one component of order greater than 1.

Observe that the edges of $\mathcal{H}$ different from the central one might be dangling. Now it remains to show that this definition of $\mathcal{H}$ enforces each cubic pole to have a reducible $\mathcal{H}$.

Lemma 4.3. *Each cubic pole of order at least 6 has a good edge.*

Proof. Each cubic pole can be constructed from K_2 and K_4 by applying a finite number of operations (a'), (b'), (c'), and (d') depicted in Figure 4.2. Observe that the extension (c) is equivalent to a consecutive application of extensions (c') and (b'). Hence, allowed are all extensions of prime poles (a) $\equiv$ (a'), (b) $\equiv$ (b'), (c) $\equiv$ (c') $\circ$ (b'), and (d) $\equiv$ (d'), together with the edge extension in (c'). As already proved in a similar lemma for cubic graphs

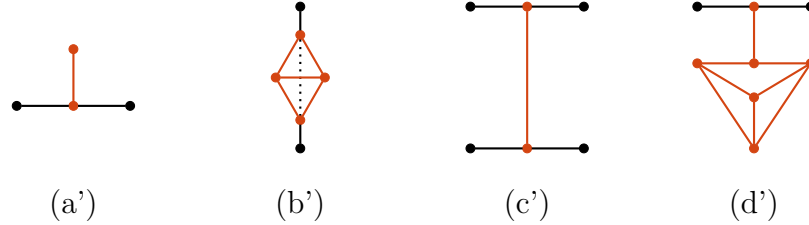


Figure 4.2: Cubic poles extensions

[3, p. 263], extensions (b'), (c'), (d') do not decrease the number of good edges. The same is true for the extension (a'), since the good edges not affected by the extension remain good, and when the affected edge is good, the two newly introduced edges are both good.

Therefore, it suffices to prove that some small poles as base cases have a good edge. However, K_2 and the tripod, as the smallest cubic poles, do not have a good edge. Hence, we need to verify larger poles.

Three poles depicted in Figure 4.3 have at least one good edge, and all edges of the K_4 are good. Moreover, one can easily check that all cubic poles on 6 and 8 vertices can be reduced to at least one of these base poles. $\square$

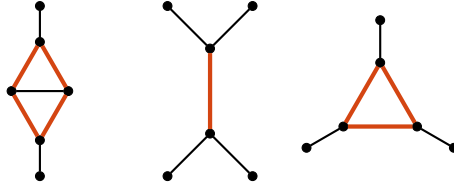


Figure 4.3: Base cases with good edges emphasised

By using this adjusted definition of a good edge in a proof of Theorem 2.5, its variant for cubic poles is a straightforward result.

Corollary 4.3.1. Each connected cubic pole P with girth $g \geq 5$ has a reducible $\mathcal{H}$. Moreover, the girth of the reduced pole is at least $g - 2$ if $g \leq 6$ and at least $g - 1$ otherwise.

4.3 Implementation challenges

We expected that the main load of the reimplementations lies in these parts:

1. a support for dangling edges – defining a list of vertex degrees, changing the iteration range from 3 to $\deg v$ where applicable, including base case of K_2 ,
2. implementation of the dangling edge extension,
3. reimplementations of the $\mathcal{H}$ operation such that some of the four affected edges may not exist.

The first two items from the previous list should be sufficient for the `polehunter` to generate all cubic poles of minimal girths 3, 4, and 5. However, after reimplementing the first two items, not all poles were generated by `polehunter`. The issue is in implicit assumptions about cubic graphs used in `snarkhunter`, not holding for cubic multipoles. The majority of them concern the invariants from Figure 2.3.

Example 1. In the following code snippet, there is a calculation of the invariant x_1 (the number of vertices in distance at most 2 from the edge) in a look-ahead before the extension of the corresponding edge. The graph G before the extension contains no triangles. The `edge_pair[0]`, `edge_pair[1]` and `edge_pair[2]`, `edge_pair[3]` are the vertex indices of the first and the second edge affected by the extension, respectively. The idea of this computation is as follows – if the local structure surrounding the new edge is a tree, due to the regularity, there are 2 endpoints of the edge in the distance zero, the four neighbours in a distance one are `edge_pair[0]`, `edge_pair[1]`, `edge_pair[2]`, `edge_pair[3]`, and each of them has exactly 2 other neighbours, which are pairwise different, so we get the desired $x_1 = 2 + 2 \cdot 2 + 2 \cdot 2 \cdot 2 = 14$. However, there

might be overlays in the neighbourhoods of these two edges. These are subtracted from the value 14.

```
edge[0] = edge_pair[0]; edge[1] = edge_pair[1];
setword neighbours0 = get_neighbours_distance_one(edge);
```

```
edge[0] = edge_pair[2]; edge[1] = edge_pair[3];
setword neighbours1 = get_neighbours_distance_one(edge);
```

```
int colour_inserted = 14 - POPC(neighbours0 & neighbours1);
```

However, this is not a correct method for cubic poles, because either of the affected edges might be dangling. This leads to a modified version of the calculation for prime poles.

```
int colour_inserted = 2 + POPC(neighbours0) + POPC(neighbours1)
    - POPC(neighbours0 & neighbours1);
```

Example 2. Another place, where the implicit consequences of the invariants play a role, is the generation of eligible edgepairs. For an illustration, consider a setting in which we know that the current graph G contains exactly two squares sharing an edge, and we expect that the extended graph has a girth strictly greater than 4 – so we need to get rid of these two squares and also do not create another one.

In general, there are two options for doing this. The first one takes an edge from each square (the choice is unique since there is only one such pair not creating a square), the second one takes the mutual edge of squares and one edge from the rest of the graph.

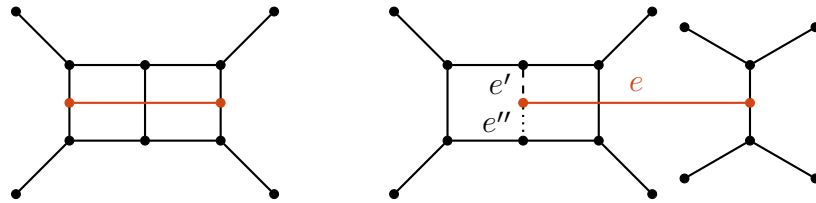
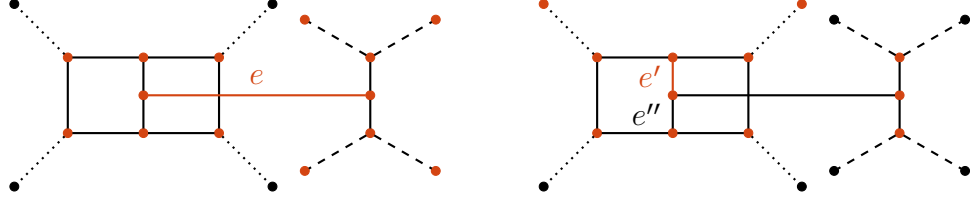


Figure 4.4: Possible extensions of a graph with exactly two squares (which are additionally adjacent)

In the original implementation of **snarkhunter**, only the left case is managed.

For cubic graphs, this is correct. Observe that in the right diagram of Figure 4.4, $x_0(e) = x_0(e') = 0$, since the extended graph from our assumption contains no squares. Now compare the values of x_1 .

From the right diagram in Figure 4.5, $x_1(e') \leq 12$. There is an inequality since some of the emphasised vertices might coincide. However, only possible collisions are when


 Figure 4.5: Neighbours in distance at most 2 from e and e'

some dashed edge coincides with some dotted edge. Moreover, when there are more than two such coinciding pairs, we can verify that it results in a cycle of length 3 or 4. Therefore, $x_1(e) \geq 14 - 2 = 12$. If $x_1(e) > x_1(e')$, the reduction of e is not canonical and we are done. Since for e'' , the figure is symmetric, $x_1(e) > x_1(e'')$ again implies that e is not canonical. So assume that $x_1(e) = x_1(e') = x_1(e'') = 12$. From $x_1(e') = 12$, no dashed edge coincide with any of the upper dotted edges. Similarly from $x_1(e'') = 12$, no dashed edge coincide with any of the lower dotted edges. Finally, no dashed edge coincide with any dotted edge, which leads to $x_1(e) = 14$, a contradiction. Therefore, the extension of e cannot be canonical.

However, the reasoning above fails for cubic multipoles, since some of the vertices in the diagram may have degree one instead of three.

Finally, we are moving to the reimplementaion of the $\mathcal{H}$ operation. The overview idea is that when generating 4-tuples of edges affected by the $\mathcal{H}$ operation (from now on, also called *edge 4-tuples*), up to three of these four edges may be flagged as not available, and the corresponding ends of the $\mathcal{H}$ will be dangling. Although this seems not as such impossible task, remind the structure of methods for generating edgepairs depicted in Figure 2.5. There is a similar structure of methods for generating edge 4-tuples, but since it affects four edges instead of only two, the number of such methods is more than twenty. This greatly increases the complexity (time, not intellectual) of the reimplementaion. Therefore, we did not implement the $\mathcal{H}$ operation, and the generator is working only for minimal girths 3, 4, and 5.

4.4 Results

The data listed in the following tables are aggregated by the number of degree 3 vertices. More granular data distinguishing the number of dangling edges can be found in the Appendix.

In Table 4.1, we summarize counts of cubic multipoles. In Table 4.2, we compare the time efficiency of `polehunter`, `multigraph`, and `pregraphs` (abbreviated as `ph`, `mg`, and `pg`, respectively).

We see that the speed up of the `polehunter` is significant, both to `pregraphs` and `multigraph`. However, from the data in Appendix, we see that the speed up is much

	All	Girth ≥ 4	Girth ≥ 5	Prime poles	Prime poles ratio [%]
1	1	1	1	1	100.00
2	1	1	1	1	100.00
3	2	1	1	1	50.00
4	6	3	2	4	66.67
5	10	5	3	4	40.00
6	29	14	6	7	24.14
7	64	29	12	10	15.62
8	194	87	29	21	10.82
9	531	219	69	36	6.78
10	1733	705	201	76	4.39
11	5524	2141	574	140	2.53
12	19430	7409	1887	297	1.53
13	69322	25727	6339	605	0.87
14	262044	96055	23093	1311	0.50
15	1016740	367294	86846	2795	0.27
16	4101318	1468938	343179	6174	0.15
17	16996157	6035982	1397699	13596	0.08
18	72556640	25611678	5889783	30542	0.04
19	317558689	111466852	25485600	68776	0.02
20	1424644848	497761656	113228666	156662	0.01

Table 4.1: Counts of cubic multipoles

	All					Girth ≥ 4			Girth ≥ 5		
	ph[s ⁻¹]	mg[s ⁻¹]	$\frac{ph}{mg}$	pg[s ⁻¹]	$\frac{ph}{pg}$	ph[s ⁻¹]	mg[s ⁻¹]	$\frac{ph}{mg}$	ph[s ⁻¹]	mg[s ⁻¹]	$\frac{ph}{mg}$
12	277571	31852	8.7	54592	5.1	148180	23153	6.4			
13	266623	32545	8.2	57768	4.6	107195	23602	4.5	63390	15461	4.1
14	272962	38994	7.0	54592	5.0	113005	29646	3.8	62413	18183	3.4
15	299041	39903	7.5	52681	5.7	121219	29958	4.0	69477	19964	3.5
16	311887	46035	6.8	51784	6.0	120701	34547	3.5	74767	22818	3.3
17	305466	44849	6.8	49065	6.2	126966	36688	3.5	78832	24738	3.2
18	322675	51393	6.3	46567	6.9	129817	40530	3.2	77938	25841	3.0
19	327502	52100	6.3	45538	7.2	129924	41706	3.1	82748	27691	3.0
20	328052					129492			83823		

Table 4.2: Running time comparison of cubic poles generators

smaller on graphs with a large number of dangling edges. Especially, the generation of acyclic poles is much faster in the **multigraph**. This is probably caused by the fact that

all acyclic poles are, by our definition, prime. Moreover, no optimisation by invariants (as in Figure 2.3) or discarding the non-feasible edgepairs (as in Figure 2.5) is applied, and so the generation of prime poles is much slower in contrast to edge extensions.

This issue can be eventually solved in two ways – either by the implementation of some optimisation for prime poles generation, or by restating the prime poles definition. For the latter case, not all dangling edges must be considered as irreducible. If we consider as irreducible dangling edges only those with a vertex in a triangle, no acyclic pole would be prime. However, this approach requires to change the prime poles operations, since current extensions are not able to generate all prime graphs without generating a non-prime one in the process.

Chapter 5

New results on $1/2$ -flow-pairs

In this chapter we show that $1/2$ -flow-pairs are worth further research, since they are in a close relation with several flow concepts. In the first part, we show their existence on snarks of oddness 2 (and hence also on weak snarks), which form the majority of currently known snarks. We also show, that $1/2$ -flow-pair implies a rich 7-flow, what is an improvement of the upper bound for all snarks of oddness 2.

5.1 Oddness 2

One of the first results on snarks admitting a NZ 5-flow was the class of snarks with oddness two (see Theorem 3.8). The main idea of the proof is adding a particular edge uv such that the resulting graph has a NZ 4-flow, and then finding an oriented uv -path, where the flow values are all positive, so we can get rid of the flow on the edge uv .

We generalise this approach and prove a similar theorem for flow-pairs. However, in this setting, we need to find two dart-disjoint uv -paths before deleting the edge uv .

Theorem 5.1. *Each snark G of oddness 2 admits a $1/2$ -flow-pair.*

Proof. Consider a 2-factor F of G with two odd cycles. Then construct an NZ $(\mathbb{Z}_2 \times \mathbb{Z}_2)$ -flow φ on G as follows – assign $(0, 1)$ to edges in $E(G) \setminus F$ and, on the 2-factor F , alternate flow values $(1, 1)$ and $(1, 0)$. Since all elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$ are involutions, assigning a value to an edge is equivalent to assigning the same value to both corresponding darts. On both odd cycles, there is one conflicting vertex, where the sum of flow values is $(0, 1)$. Let these conflicting vertices be u and v . Join these two vertices by a new edge e with an assigned flow value $(1, 1)$ and denote the new graph by G' . Thus, we have an NZ $(\mathbb{Z}_2 \times \mathbb{Z}_2)$ -flow φ on G' . By Theorem 3.2 we transform this NZ $(\mathbb{Z}_2 \times \mathbb{Z}_2)$ -flow to a $0/1$ -flow-pair. However, we know that $\varphi(\overrightarrow{uv})$ is either $(0, 1)$ or $(0, -1)$. In the latter case, we can multiply the second coordinate of the whole flow by -1 . Therefore we can WLOG assume $\varphi(\overrightarrow{uv}) = (0, 1)$.

Multiply the second coordinate of φ by two to obtain $\varphi' = (\varphi'_1, \varphi'_2) = (\varphi_1, 2\varphi_2)$. Observe that the flow values of φ' are feasible for a 1/2-flow-pair and $\varphi'(\vec{uv}) = (0, 2)$. Call a dart d *good* if the value of $\varphi'(d) + (0, 1)$ is still a feasible flow value of the 1/2-flow-pair. Now, if we find two oriented cycles containing the dart $\vec{vu}$ such that all other darts in cycles are good, we can get rid of the flow on the edge uv and then delete it. Clearly, we cannot take the same dart twice, but when we take two darts corresponding to the same edge in G and send $(0, 1)$ through each of them, the resulting flow value on e is the same as before the process, so this is no issue. Now, we construct a graph H of good darts, formally $V(H) = V(G)$ and

$$E(H) = \{\vec{xy} \mid xy \in E(G) \wedge (\varphi'_1(\vec{xy}) = \pm 1 \vee \varphi'_2(\vec{xy}) = 2)\}.$$

Now we aim to find two dart-disjoint uv -paths in H . From Theorem 1.1, there exist two dart-disjoint $\vec{uv}$ -paths in H if and only if each uv -dart-separator of H contains at least 2 darts from the component containing u to the component containing v . So it remains to prove this condition.

Consider any uv -dart-separator $\gamma_{G'}$ of G' . From the generalised flow conservation, $\sum_{d \in \gamma_{G'}} \varphi'(d) = (0, 0)$. If there is a dart in $\gamma_{G'}$ having a flow value with non-zero first coordinate, there must be at least one more such dart due to the sum of flow values. For both of them, there exists a corresponding dart in H . Hence, we have at least two darts in the corresponding uv -dart-separator γ_H of H . In the remaining case, only possible flow values on the darts of $\gamma_{G'}$ are $(0, \pm 2)$.

Observe that the number of darts with flow value $(0, 2)$ must be the same as the number of darts with flow value $(0, -2)$ due to the zero sum. This also argues that $|\gamma_{G'}|$ is an even number. One of the $(0, 2)$ darts is the new uv edge, and for the others, there is a corresponding dart in γ_H . Hence, if $|\gamma_{G'}| > 4$, there are at least three $(0, 2)$ darts in $\gamma_{G'}$, and so at least two $(0, 2)$ darts in γ_H . If $|\gamma_{G'}| = 2$, there is a bridge in G , which contradicts having any NZ flow. So we are left with $|\gamma_{G'}| = 4$.

From $|\gamma_{G'}| = 4$ we already know that $|\gamma_G| = 3$. Since G as a snark is cyclically 4-edge-connected, at least one of the components separated by γ_G must be an acyclic 3-pole with k inner vertices. Let k denote the number of its inner vertices of degree 3. We will use the fact that this 3-pole is a tree T on $k + 3$ vertices with 3 leaves, and hence $|E(T)| = k + 2$. Moreover, from the folklore handshaking lemma, we get

$$3k + 3 = \sum_{v \in V(T)} \deg(v) = 2|E(T)| = 2k + 4,$$

and so $k = 1$. However, the only cases when the uv -dart-separator separates one vertex, are the separators around u and v . Both of these separators contain a flow value with non-zero first coordinate, so this case is already solved. $\square$

Corollary 5.1.1. Each weak snark admits a 1/2-flow-pair.

5.2 Rich 7-flow

Now we move to the relation of rich flows and flow-pairs. Up to now, the smallest upper bound proved for some infinite family of snarks is in the Proposition 3.23, which shows that if we can delete an edge, smooth the vertices of degree 2 and obtain a colourable graph, the original graph has the rich flow number at most 9. However, from the 1/2-flow-pair, we are able to construct a rich NZ 7-flow. This means, that for graphs with oddness 2, we tightened the gap between the construction and the bound from 3 to only 1.

Theorem 5.2. *Let G be a cubic graph with 1/2-flow-pair. Then G has a rich nowhere-zero 7-flow.*

Proof. Denote the 1/2-flow-pair of G as $\varphi = (\varphi_1, \varphi_2)$. First, we construct a circular nowhere-zero 7-flow φ' . Then, we will adjust non-integral values of φ' to φ'' , which will be the sought rich nowhere-zero 7-flow.

Let $\varphi'(d) = 9\varphi_1(d)/2 + \varphi_2(d)/2$. We see that

$$\begin{aligned} \varphi_1(d) = -1 & \Rightarrow \varphi'(d) \in [-6, -3], \\ \varphi_1(d) = 0 & \Rightarrow \varphi'(d) \in \{-3/2, -1, 1, 3/2\}, \\ \varphi_1(d) = 1 & \Rightarrow \varphi'(d) \in [3, 6]. \end{aligned}$$

This shows that φ' is a circular nowhere-zero 7-flow.

From Proposition 3.12, we can construct an integral nowhere-zero 7-flow φ'' . Moreover, its flow values satisfy

$$\begin{aligned} \varphi_1(d) = -1 & \Rightarrow \varphi''(d) \in \{-6, -5, -4, -3\}, \\ \varphi_1(d) = 0 & \Rightarrow \varphi''(d) \in \{-2, -1, 1, 2\}, \\ \varphi_1(d) = 1 & \Rightarrow \varphi''(d) \in \{3, 4, 5, 6\}. \end{aligned}$$

This is the key observation to prove that φ'' is rich. For the sake of contradiction, assume a vertex v , in which two darts are either confluent or contrafluent. They cannot be confluent, since otherwise the third edge would have zero flow value. Hence, denote the common value on the contrafluent darts as x and on the third dart, there is $-2x$, which is in absolute value greater. Hence, the two equal flow values must have the smallest absolute value. From Lemma 3.19, each vertex is surrounded by one dart with $\varphi_1(d) = 0$ and two darts with $\varphi_1(d) = 1$. But absolute values of $\varphi''(d)$ on darts with $\varphi_1(d) = 0$ are strictly smaller than absolute values of $\varphi''(d)$ on darts with $\varphi_1(d) \neq 0$. Therefore, there cannot be two darts with the same smallest absolute value, a contradiction. Finally, we get that φ'' is a rich NZ 7-flow. $\square$

Corollary 5.2.1. Each snark G of oddness 2 has a rich NZ 7-flow.

Chapter 6

Chebyshev flow number of snarks

In this chapter, we prove a lower bound on the Chebyshev flow number of snarks depending on its order, with 5 examples achieving the equality. Although we do not find an infinite class of snarks proving this bound is tight, we construct a class of graphs, where only two vertices are of degree different from 3, where their flow number corresponds to the bound for the order of this graph. Hence, these graphs might be good starting points for constructing the infinite class attaining the bound. Moreover, we show that these graphs can be adjusted to snarks with a smaller flow number, and hence, a snark with arbitrarily small flow number exists. Finally we construct a Chebyshev flow on generalised Blanuša snarks.

6.1 Lower bound

In one-dimensional flows, there has been proved a lower bound on the flow number of a snark

$$\Phi_1(G) \geq 4 + 1 \left\lceil \frac{n-4}{8} \right\rceil$$

depending on the number n of its vertices [21, p. 14]. Moreover, there is also a meaningful lower bound on the two-dimensional Chebyshev flow number of the Petersen graph P [32, p. 99]. It states that $\Phi_2^\infty(P) \geq 5/2$. We provide a generalisation of this bound for any snark. Additionally, the bounding number is roughly a half of the bounding number in one dimension.

Proposition 6.1. [10, pp. 15, 16] *Let G denote a snark of order n . Then $\Phi_2^\infty(G) \geq 2 + 1/\xi$, where $\xi = \left\lfloor \frac{n-2}{4} \right\rfloor$.*

Proof. For the sake of contradiction assume that G admits a 2D Chebyshev NZ flow φ such that each edge $e \in E(G)$ satisfies $\varphi(e) = (\varphi_1(e), \varphi_2(e))$, $\|\varphi(e)\|_\infty \geq 1$, and $|\varphi_i(e)| < 1 + 1/\xi$ for $i \in \{1, 2\}$.

We call an edge $e \in E(G)$ *large* with respect to φ_i if $|\varphi_i(e)| \geq 1$, and *small* otherwise. Observe that an edge e can be large with respect to both φ_1 and φ_2 , but it cannot be small with respect to both φ_1 and φ_2 , since otherwise $\|\varphi(e)\|_\infty < 1$.

For $i \in \{1, 2\}$, denote by S_i and L_i the subgraphs of G induced by the small and large edges with respect to φ_i , respectively. WLOG $|S_1| \leq |S_2|$. Moreover, from the previous observation, S_1, S_2 are disjoint, and hence $|S_1| \leq \lfloor \frac{1}{2}|E(G)| \rfloor = \lfloor \frac{3n}{4} \rfloor$.

Claim 1. S_1 contains no vertex of degree 3.

Proof. For the sake of contradiction, assume a vertex v surrounded by three small edges w.r.t. 1. Then, vertex v must be surrounded also by three large edges w.r.t. 2. Denote by d_1, d_2, d_3 the darts going out of v . WLOG $\varphi_2(d_1) \geq \varphi_2(d_2) \geq 0 \geq \varphi_2(d_3)$. Then, the flow conservation requires

$$0 = \varphi_2(d_1) + \varphi_2(d_2) + \varphi_2(d_3) \geq 1 + 1 + (-1 - 1/\xi) = 1 - 1/\xi,$$

which is equivalent to $\xi \leq 1$, and hence $n \leq 9$. But there are no snarks with less than 10 vertices, a contradiction. ■

Claim 2. If $C \subseteq E(L_i)$ is an odd edge-cut of G for any $i \in \{1, 2\}$, then $|C| \geq 2\xi + 3$.

Proof. Consider an odd edge-cut C of G containing $2k+1$ large edges w.r.t. i separating components G_1, G_2 . Denote the corresponding darts from G_1 to G_2 as $d_1, d_2, \dots, d_{2k+1}$. WLOG assume that $\varphi_i(d_j) \geq 0$ for all $1 \leq j \leq k+1$. Then, the generalised flow conservation requires that

$$0 = \sum_{j=1}^{2k+1} \varphi_i(d_j) = \sum_{j=1}^{k+1} \varphi_i(d_j) + \sum_{j=k+2}^{2k+1} \varphi_i(d_j) > (k+1) \cdot 1 + k \cdot (-1 - 1/\xi) = 1 - k/\xi,$$

which is equivalent to $k > \xi$. ■

Claim 3. A path of length $2k$ with $k < \xi$ cannot be a component of S_1 .

Proof. For the sake of contradiction assume that S_1 contains a path of length $2k$ with $k < \xi$. Then the edges adjacent to this path are in L_1 and they form an odd edge-cut of G , containing at most $2k+3 < 2\xi+3$ edges, which is in contradiction with Claim 2. ■

Claim 4. A cycle of length $2k+1$ with $k \leq \xi$ cannot be a component of S_1 .

Proof. For the sake of contradiction assume that S_1 contains a cycle of length $2k+1$ with $k \leq \xi$. Then the edges adjacent to this cycle are in L_1 and they form an odd edge-cut of G , containing at most $2k+1 \leq 2\xi+1 < 2\xi+3$ edges, which is in contradiction with Claim 2. ■

Claim 5. S_1 cannot contain a perfect matching of G .

Proof. For the sake of contradiction assume that S_1 contains a perfect matching M of G . Then also L_2 contains M . Next, $F = E(G) \setminus M$ is a 2-factor of G . Since G is a snark, its oddness is at least two, and therefore, F contains at least two odd cycles. Then, the smallest one has length $2k + 1 \leq \frac{n}{2}$. It follows that $k \leq \frac{n-2}{4}$, and since k is natural, we also have $k \leq \lfloor \frac{n-2}{4} \rfloor = \xi$. Then the edges adjacent to this cycle are in L_2 , and they form an odd edge-cut of G , containing at most $2k + 1 \leq 2\xi + 1$ edges, which is in contradiction with Claim 2. ■

By Claim 1 we know that S_1 contains only paths and cycles. Note that by Claim 3, each even path in S_1 contains at least 2ξ edges. Similarly by Claim 4, each odd cycle in S_1 contains at least $2\xi + 3$ edges. Let *odd components* denote odd cycles and even paths. We aim to show that S_1 has less than two odd components. Otherwise,

$$\left\lfloor \frac{3n}{4} \right\rfloor \geq E(S_1) \geq 4\xi = 4 \left\lfloor \frac{n-2}{4} \right\rfloor. \quad (6.1)$$

We disprove this inequality by induction. For $n \in \{10, 20\}$ clearly $\left\lfloor \frac{3n}{4} \right\rfloor < 4 \left\lfloor \frac{n-2}{4} \right\rfloor$. Increasing the order n by 4 results in increasing the LHS and RHS by 3 and 4, respectively. Hence, the (6.1) holds for no even $n \geq 10$ except 12 and 16. However, there are no snarks of order 12, 16 or smaller than 10, a contradiction.

Therefore, S_1 contains at most one odd component. On the other hand, S_1 has even order, so it contains even number of odd components. As a result, S_1 contains no odd component, so it consists only of odd paths and even cycles. These also induce a perfect matching of G , but that is in contradiction with Claim 5. □

For now, we are aware of five snarks meeting this bound – Petersen graph P with the flow number $5/2$, both Blanuša snarks $B_2^{(1)}, B_2^{(2)}$ with flow numbers $9/4$ (which is a part of more general result in Proposition 6.6) and 2 out of 6 snarks on 20 vertices with flow numbers also equal to $9/4$. From computational results, we know that no other example of tightness exists on at most 26 vertices [10, p. 17]. Although we do not know about any other snark meeting this bound, we think the bound is not so far from reality, since we found an infinite class of graphs that are nearly cubic and meet this bound.

6.2 Petersen book graphs

The mentioned class was already introduced when constructing graphs with given circular flow number [27, p. 312].

Definition. Let Q be the Petersen graph without an edge (since the Petersen graph is edge-transitive, it does not matter which one is removed) and denote its two vertices of degree two by x and y . Then, a *Petersen book* PB_n is a graph, which arises from

n copies of Q by identifying all vertices corresponding to x in a vertex $\hat{x}$, similarly all vertices corresponding to y in $\hat{y}$ and finally joining $\hat{x}$ and $\hat{y}$ by an edge.

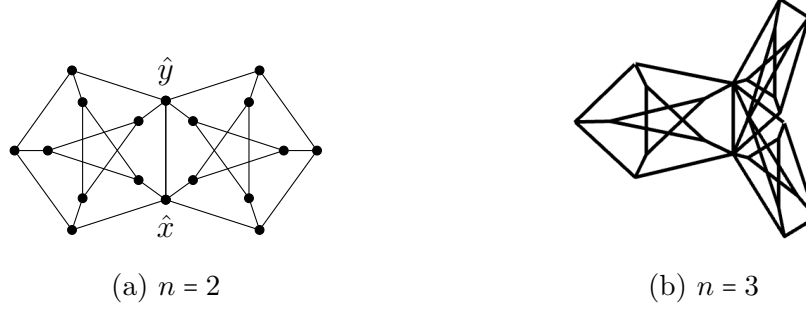


Figure 6.1: Two small examples of PB_n graphs

Lemma 6.2. [27, p. 312] *For each positive integer n , $\Phi_1(PB_n) = 4 + 1/n$.*

From Proposition 3.17, one can easily see, that if PB_n admits a t -flow-pair, then $t \geq 1/(2n)$. And for $t = 1/(2n)$, the flow-pair also exists.

Proposition 6.3. *For each positive integer n , PB_n admits a $(1/(2n))$ -flow-pair.*

Proof. In Figure 6.2, we define flow values of flow-pair (φ_1, φ_2) on one copy of Q . The edges of the 2-factor induced by φ_1 are in bold. We can check that in each vertex

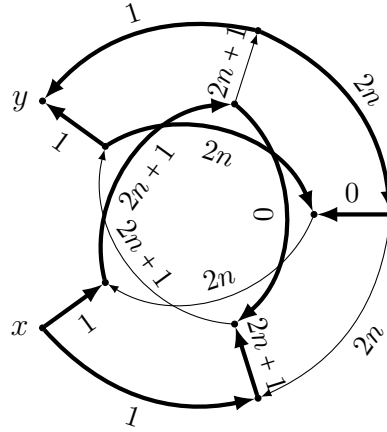


Figure 6.2: $(1/(2n))$ -flow-pair on Q

except x and y , the flow conservation constraint is satisfied, and non-bolded edges use only flow values $2n$ and $2n+1$. Hence, all flow values used in one copy of Q are feasible. Moreover, when we join n copies of Q , the sum of flow values outgoing from $\hat{x}$ will be $(0, 2n)$, and the same will be the sum of flow values ingoing to $\hat{y}$. However, this is also a feasible flow value, so by setting $\varphi(\vec{\hat{y}\hat{x}}) = (0, 2n)$, we get the desired $1/(2n)$ -flow-pair of PB_n . $\square$

This Proposition also gives an upper bound on the Chebyshev flow number of PB_n .

Corollary 6.3.1. For each positive integer n , $\Phi_2^\infty(PB_n) \leq 2 + 1/(2n)$.

By a computer, we verified that $\Phi_2^\infty(PB_3) = 2 + 1/6$ and $\Phi_2^\infty(PB_4) = 2 + 1/8$, so we expect that the inequality can be replaced by the equality.

Conjecture 6.4. For each positive integer n , $\Phi_2^\infty(PB_n) = 2 + 1/(2n)$.

Note that PB_n is of order $8n + 2$. Even though PB_n is not cubic for $n \geq 2$ (due to two vertices of degree $2n + 1$), observe that by inserting the order into the bound from Proposition 6.1, we get $1/(2n)$. Therefore, if we were able to split each of the two vertices of degree $2n + 1$ into vertices of degree 2 and one vertex of degree 3 without increasing the flow number or creating a 3-edge-colourable graph, after smoothing vertices of degree 2, we will obtain a graph meeting the lower bound.

This is possible for $n = 2$, where we can obtain both Blanuša snarks. For $n = 3$, we already know this is not possible, because all snarks on 26 vertices have 2D Chebyshev flow number at least $2 + 1/4$. However, we are able to reduce the degrees of these vertices so that in the resulting graph, only one vertex is not of degree 3, and its degree is 5. Then we replaced this vertex by a 5-cycle. By computer, we verified that the flow number still remains the same, and so we constructed a snark on 30 vertices with 2D Chebyshev flow number $2 + 1/6$. Hence, we suspect, no snark with 2D Chebyshev flow number $2 + 1/6$ on 28 vertices exists.

In summary, adjusting PB_n without changing the flow number can lead to snarks with Chebyshev flow number near the proved bound. Now we will show one general construction, giving an upper bound on the order of the graph with sufficiently small flow number. This method is inspired by a similar construction for circular flow number [22, pp. 193-195].

Proposition 6.5. Let G be a graph with a p/q -flow-pair. Then there exists a cubic graph G' with a p/q -flow-pair.

Proof. Fix a p/q -flow-pair of G and consider a vertex v with degree d greater than 3. We aim to construct a cubic d -pole, which will be the replacement of v in G . Therefore, the multiset of the flow values on dangling edges of this pole should be the same as the multiset of flow values surrounding v in the original graph.

Let us start with the d dangling edges with corresponding flow values. Now, we attach graphs to the dangling edges, to obtain only dangling edges with flow values $(0, \pm(p+q)), (\pm 1, 0)$ in the end. To get rid of unwanted edges with zero first coordinate and $a > 0$, we will use the graph in Figure 6.3. The graph for $a < 0$ can be obtained in a similar manner.

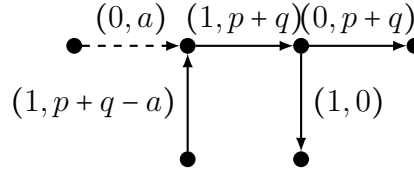


Figure 6.3

So now we are left with the unwanted edges with a non-zero first coordinate. Take two dangling edges with first coordinates equal to 1 and second coordinates $-p - q < a \leq b < p + q$. Let $x = \min\{p + q - b, p + q + a\}$. We attach $\lfloor x/p \rfloor$ copies of the graph in Figure 6.4 with $\delta = p$ and then another copy with $\delta = x \bmod p$. This causes we get either $(1, -p - q)$ or $(1, p + q)$ on one of the dangling edges. To this dangling edge, we can attach dangling edges with flow values $(1, 0)$ and $(0, \pm(p + q))$, and hence reduce the number of unwanted edges. Since this can be done while there are at least two dangling edges with the first coordinate equal to 1, we will end up with at most one conflicting edge, whose first coordinate of flow value is 1. By a similar reasoning, we can deal with dangling edges, where the first coordinate of the flow value is -1 .

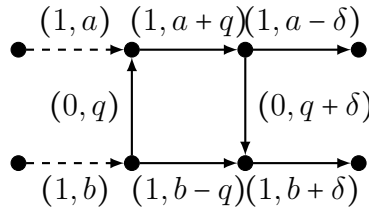


Figure 6.4

At this moment, the majority of dangling edges have flow values $(\pm 1, 0)$ and $(0, \pm(p + q))$. The only two possible exceptions are $(1, s)$ and $(-1, t)$ for some $-p - q < s, t < p + q$. From the flow conservation constraint, we deduce that $s + t \equiv 0 \pmod{p + q}$. If $s + t = 0$, we can attach those dangling edges and smooth their common vertex of degree 2. On the other hand, if $s + t = \pm(p + q)$, we will additionally attach a $(0, \pm(p + q))$ dangling edge to fulfill the flow conservation constraint. As the last step, we realize, that from the flow conservation constraint, the number of $(1, 0)$ dangling edges is equal to the number of $(-1, 0)$ dangling edges (and same for the $(0, \pm(p + q))$ edges), so we can attach edges with opposite flow values and smooth their common vertex. Finally, we ended up with (not necessarily connected) cubic d -pole with desired properties, and hence, we constructed a cubic graph with p/q -flow-pair. $\square$

Corollary 6.5.1. For any $r > 2$, there exists a non-colourable cubic graph whose 2D Chebyshev flow number is at most r .

6.3 Generalised Blanuša snarks

Now, we move to generalised Blanuša snarks. In one dimension, the flow number of generalised Blanuša snarks is proved to be $4 + 1/2$. The paper concerning these graphs also shows an explicit construction of the circular flow [20, p. 978]. We modify this construction and construct a $1/4$ -flow-pair. From this, we know that all generalised Blanuša snarks have the 2D Chebyshev flow number at most $2 + 1/4$.

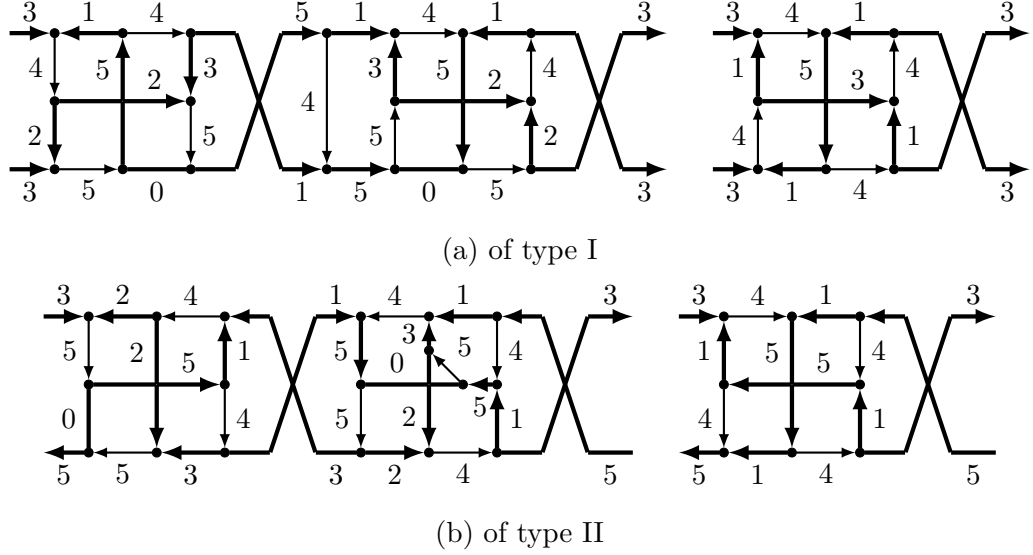


Figure 6.5: A $1/4$ -flow-pair on generalised Blanuša snarks

Proposition 6.6. *For each $n \geq 2$, both $B_n^{(1)}$ and $B_n^{(2)}$ have a ChNZ $9/4$ -flow.*

Proof. In Figures 6.5a and 6.5b, we illustrate the $1/4$ -flow-pairs $\varphi = (\varphi_1, \varphi_2)$ on generalised Blanuša snarks, where the edges of the 2-factor induced by φ_1 are in bold. Then, from Proposition 3.17, these graphs have a ChNZ $9/4$ -flow. $\square$

From Proposition 6.1, we know that $\Phi_2^\infty(B_2^{(1)}) = \Phi_2^\infty(B_2^{(2)}) \geq 9/4$, so for the two Blanuša snarks, the bound meets the construction. For generalised Blanuša snarks on 26 and 34 vertices, we verified by computer that their 2D Chebyshev flow number is $9/4$. Therefore, we suspect it holds in general.

Conjecture 6.7. *For each $n \geq 2$, $\Phi_2^\infty(B_n^{(1)}) = \Phi_2^\infty(B_n^{(2)}) = 9/4$.*

Conclusion

In Chapter 4, we developed a generator of cubic multipoles `polehunter` with minimum girths 3, 4 and 5. This algorithm is significantly faster than other known generators `multigraph` and `pregraphs`, and by a theoretical background, we show a possibility to extend the algorithm to minimal girths 6 and 7.

In Chapter 5, we proved that graphs of oddness 2 have a $1/2$ -flow-pair and also graphs with a $1/2$ -flow-pair admit a rich NZ 7-flow. Hence, we used the $1/2$ -flow-pair as the “structure-in-the-middle”. We believe that the $1/2$ -flow-pair can be used to prove more results in the NZ flow theory. For example, when someone proves the existence of the $1/2$ -flow-pair for some uncolourable class of snarks, he “kills three birds with one stone”, since it automatically implies the NZ 5-flow, the ChNZ $5/2$ -flow, and also the rich NZ 7-flow. In a similar way, when someone shows a construction of some flow from the $1/2$ -flow-pair, it is immediately proved for all classes implying $1/2$ -flow-pair.

In Chapter 6, we proved a lower bound on two-dimensional Chebyshev flow number of snarks. Although we are not aware of any infinite class meeting this bound, we described the infinite class of so-called *Petersen book* graphs, which are not cubic, but their flow number corresponds to the lower bound for snarks of same order. Then we constructed snarks with order higher than the order of Petersen books, but with the flow number that is not greater than the flow number of Petersen books. This shows that the two-dimensional Chebyshev flow number can be arbitrarily near to 2 for snarks.

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Appendix: Data concerning polehunter

Counts girth 3

	0	1	2	3	4	5	6	≥ 7	Total
0			1						1
1				1					1
2					1				1
3				1		1			2
4	1		1		2		2		6
5		1		3		4		2	10
6	2		5		9		9	4	29
7		4		15		22		23	64
8	5		23		54		63	49	194
9		19		91		166		255	531
10	19		132		390		551	641	1733
11		114		671		1492		3247	5524
12	85		949		3433		5741	9222	19430
13		839		6073		15773		46637	69322
14	509		8428		36245		69602	147560	262344
15		7618		65451		193699		749972	1016740
16	4060		90562		447641		965223	2593832	4101318
17		83134		821162		2711487		13380374	16996157

Running times ratio girth 3

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
13							7		6		3						
14				7		7		7		8		7		2		< 1	
15			12		7.5		9.8		7.7		6.6		6		3		0.5
16		7.5		8.5		5.9		8.6		6.8		5.8		4.8		3.5	
17	7		8.6		6.9		8.4		7.2		5.8		5.6		4.6		3.1

Counts girth 4

	0	1	2	3	4	5	6	≥ 7	Total
0			1						1
1				1					1
2					1				1
3				1		1			2
4	1		1		2		2		6
5		1		3		4		2	10
6	2		5		9		9	4	29
7		4		15		22		23	64
8	5		23		54		63	49	194
9		19		91		166		255	531
10	19		132		390		551	641	1733
11		114		671		1492		3247	5524
12	85		949		3433		5741	9222	19430
13		839		6073		15773		46637	69322
14	509		8428		36245		69602	147560	262344
15		7618		65451		193699		749972	1016740
16	4060		90562		447641		965223	2593832	4101318
17		83134		821162		2711487		13380374	16996157

Running times ratio girth 3

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
13							7		6		3						
14				7		7		7		8		7		2		< 1	
15			12		7.5		9.8		7.7		6.6		6		3		0.5
16		7.5		8.5		5.9		8.6		6.8		5.8		4.8		3.5	
17	7		8.6		6.9		8.4		7.2		5.8		5.6		4.6		3.1

Girth 3

$d_3 \backslash d_1$	0	2	4	6	8	10	12	14	16	18	20
2											
4											
6											
8											
10											
12			9	11	10	11	5	1			
14		9	10	7	9	8	6	4	0		
16	9	10	8	6	9	7	6	5	3	0	
18	8	10	7	6	8	6	6	5	4	3	0

	0	2	4	6	8	10	12	14	16	18	20	22
2	0	0	1									
4	1	1	2	2								
6	2	5	9	9	4							
8	5	23	54	63	38	11						
10	19	132	390	551	424	180	37					
12	85	949	3433	5741	5291	2888	908	135				
14	509	8428	36245	69602	74116	48182	19644	4766	552			
16	4060	90562	447641	965223	1155398	859634	417826	132776	25788	2410		
18	41301	1140814	6314647	15027908	19866458	16491723	9152744	3477106	890648	142271	11020	
20	510489	16383987	99745375	258851111	373236848	340064489	210004798	91097327	27974520	5927900	795771	52233
	1	3	5	7	9	11	13	15	17	19	21	
3	0	1	1									
5	1	3	4	2								
7	4	15	22	17	6							
9	19	91	166	157	80	18						
11	114	671	1492	1694	1092	395	66					
13	839	6073	15773	20818	16005	7494	2055	265				
15	7618	65451	193699	289254	255089	142469	51004	11024	1132			
17	83134	821162	2711487	4495568	4434775	2830324	1210300	343942	60367	5098		
19	1059165	11718309	42524910	77224877	83808081	59578229	29087790	9898531	2299170	335781	23846	

$d_3 \backslash d_1$	1	3	5	7	9	11	13	15	17	19	21
3											
5											
7											
9											
11											
13		13	11	12	8	6	5	1			
15	16	10	8	10	8	7	5	4	0		
17	10	9	8	8	7	6	6	4	3	0	
19	10	9	7	8	7	6	5	5	4	3	0

Girth 4

$d_3 \backslash d_1$	0	2	4	6	8	10	12	14	16
2	0	0	1						
4	0	0	1	2					
6	1	1	3	5	4				
8	2	6	17	27	24	11			
10	6	32	111	198	201	120	37		
12	22	213	910	1862	2138	1496	633	135	
14	110	1818	9279	21300	27188	21456	10906	3446	552
16	792	19464	113522	286515	399830	348296	201033	77872	19204
18	7805	248838	1604020	4388407	6625820	6282165	4010092	1779064	545920

$d_3 \backslash d_1$	1	3	5	7	9	11	13	15			
3	0	0	1								
5	0	1	2	2							
7	1	4	8	10	6						
9	4	23	51	65	48	18					
11	26	173	465	648	553	275	66				
13	215	1635	5005	7827	7147	4105	1537	265			
15	2199	19489	66055	112412	113382	73395	30874	8878			
17	26824	280598	1049584	1943158	2168087	1565996	775369	266110			
19	380224	4425001	18188631	36794143	44982225	35921379	19789264	7988069			
$d_3 \backslash d_1$	0	2	4	6	8	10	12	14	16	18	20
2											
4											
6											
8											
10											
12			4.20	4.50	4.40	3.90	2.60	1.00			
14		4.00	4.50	3.90	4.00	3.60	2.90	2.30	0.40		
16	4.00	4.20	4.30	3.70	4.10	3.50	3.00	2.80	2.00	0.40	
18	4.00	4.50	3.90	3.50	3.90	3.40	3.00	2.80	2.70	2.00	0.30
$d_3 \backslash d_1$	1	3	5	7	9	11	13	15	17	19	21
3											
5											
7											
9											
11											
13		5.00	4.90	4.40	4.20	3.40	2.50	0.60			
15	5.80	4.50	3.90	4.10	3.80	3.40	3.00	2.00	0.40		
17	4.50	4.30	4.00	4.00	3.70	3.40	3.10	2.70	2.00	0.40	
19	4.60	4.50	3.70	3.80	3.70	3.20	3.00	2.90	2.70	2.00	0.30

Girth 5

$d_3 \backslash d_1$	0	2	4	6	8	10	12	14	16	
2	0	0	1							
4	0	0	1	2						
6	0	0	2	4	4					
8	0	2	9	20	22	11				
10	1	8	44	110	145	102	37			
12	2	49	291	773	1080	894	501	135		
14	9	352	2542	7898	12352	11486	7189	2910	552	
16	53	3620	30457	105533	186359	190352	136927	61948	16100	24
18	410	45009	435061	1663147	3283799	3651965	2946787	1506658	480198	1010

$d_3 \backslash d_1$	1	3	5	7	9	11	13	15	17	19
3	0	0	1							
5	0	0	1	2						
7	0	1	5	8	6					
9	1	9	31	52	46	18				
11	8	77	274	493	485	260	66			
13	79	793	3167	5599	5665	3746	1426	26		
15	952	10644	44891	88057	98935	73036	35174	779		
17	13571	170449	794894	1638520	2015510	1625888	911242	33878		
19	224845	3126427	16069595	34016483	45546992	40257785	24977104	1156452		

$d_3 \backslash d_1$	0	2	4	6	8	10	12	14	16	18	20
2											
4											
6											
8											
10											
12			2.20	2.20	2.20	1.80	1.80	1.00			
14		2.20	2.40	2.40	2.30	2.10	1.90	1.70	0.40		
16	2.00	2.40	2.50	2.30	2.40	2.10	2.00	1.90	1.50	0.40	
18	2.00	2.40	2.40	2.30	2.30	2.10	2.00	1.90	1.90	1.50	0.30

$d_3 \backslash d_1$	1	3	5	7	9	11	13	15	17	19	21
3											
5											
7											
9											
11											
13		2.50	2.40	2.40	2.30	2.00	1.70	0.60			
15	3.00	2.50	2.40	2.40	2.30	2.10	2.00	1.50	0.40		
17	2.50	2.50	2.40	2.30	2.30	2.20	2.00	1.90	1.50	0.40	
19	2.60	2.50	2.30	2.30	2.30	2.10	2.00	1.90	1.90	1.50	0.30